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Palaeomagnetism and age of mid-Palaeozoic ring complexes in Niger, West Africa, and tectonic implications

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Summary. Detailed AF and thermal demagnetization has been performed on 185 oriented samples from 36 sites in a series of basic to acid subvolcanic ring complexes in central and southern Niger. Published Rb–Sr whole rock and mineral isochrons from acid members of the suite range – north to south – from 480 Ma to 320 Ma.

Two distinct vectors have been isolated by linear regression of demagnetization data:

(1) A due north and negative inclination (rarely reversed) component is present in basic rocks from 12 sites in four distinct northern complexes (published Rb–Sr ages date from 480 Ma to 420 Ma). This component is carried by ~single domain magnetite (occasionally also by titanhaematite) and is considered to be primary, acquired during the terminal stages of this igneous episode. $^{40}Ar/^{39}Ar$ dating of biotite from three samples used in the palaeomagnetic study yield ~435 Myr ages.

(2) A due north and positive inclination (rarely reversed) direction is present as a second component in some of the basic-suite sites and is the predominant component in most sites in syenitic to granodioritic rocks. Primary Fe–Ti oxides are rare in these acid igneous rocks, and the feldspars within them are usually pervasively altered. This component, carried primarily by haematite, is believed to be either a modern day VRM, or a secondary TCRM acquired during (hydro?) thermal activity associated with widespread Quaternary volcanism in the southern Air.

The primary component gives a mean south pole at 43.4S, 08.6E; ($n = 12$, $k = 50$, $\alpha_{95} = 6.2$) located southwest of Cape Town. This position requires a rapid southward excursion of the Gondwana A.P.W. path in early Silurian time, but it is generally consistent with most of the other Silurian/Devonian palaeomagnetic data from Gondwana. These, however, are from

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the Lachlan Fold Belt in southeast Australia whose relation to the Australian craton in Early Palaeozoic time is not clear.

Integrating all stratigraphic, palaeoclimatic, tectonic, and other palaeomagnetic data for Early and Middle Palaeozoic time, suggests that the proto-Atlantic closed during the Silurian Period. If a late Devonian–early Carboniferous Gondwana pole in central Africa is valid, the ocean may have opened again before the final closure and reassembly of Gondwana in the late Carboniferous.

Key words: dating, Niger, palaeomagnetism, ring-complexes, Mid-Palaeozoic

Introduction

A series of granitic ring complexes, occurring in three main groups, extend meridionally over 1400 km across Niger and Nigeria, West Africa, (Fig. 1), and range in age from ~480 Myr in

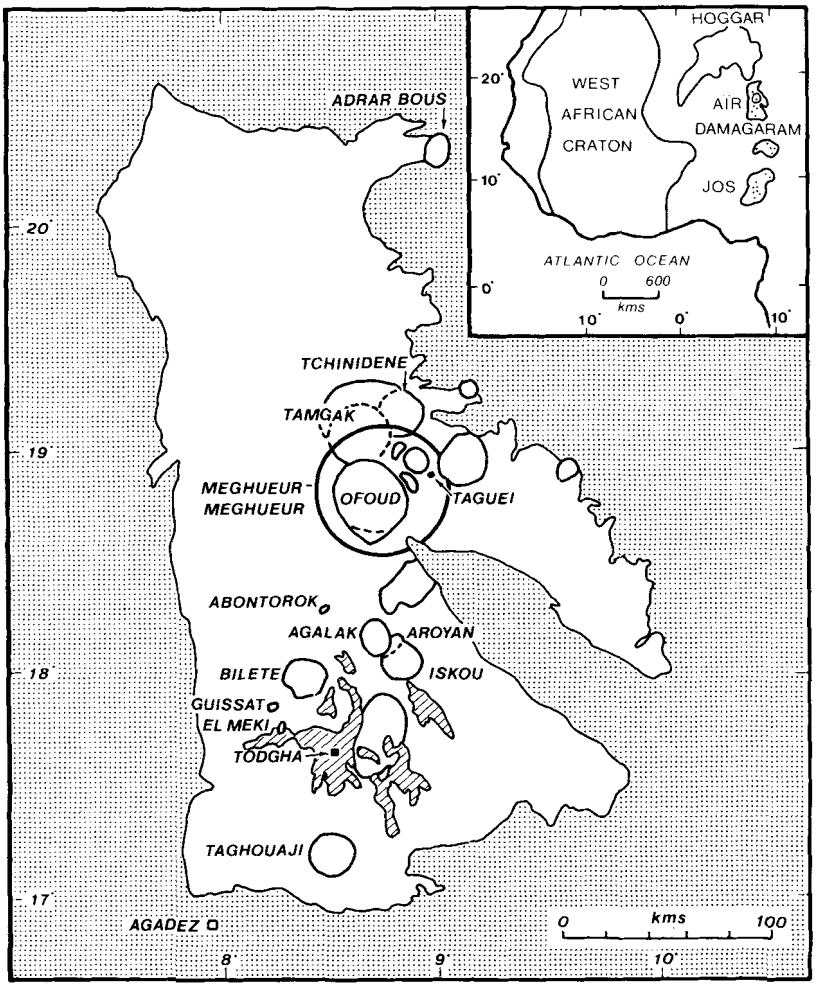


Figure 1. Map of Air Massif showing outlines of ring complexes and localities referred to in text. Inset shows location of Air, Damagaram, and Jos Plateau in West Africa.

the north, to ~180 Myr in Nigeria (Karche & Vachette 1976, 1978; Bowden *et al.* 1976). Several of the intrusive complexes in the northernmost group, in the Air (pronounced 'eye-ear') region of Niger, contain a variety of anorthositic rocks (Black, Jaujou & Pellaton 1967; Black & Girod 1970). In the course of a petrological study of some of these rocks (Husch & Moreau 1982; Husch 1982; Moreau 1982) oriented samples for palaeomagnetic study were collected in 1977, 1978 and 1979 from complexes throughout Niger (480 Ma to 300 Ma).

Sampling and measurement

One hundred and seventy-five hand samples, oriented with a sun compass, were collected from 34 sites (Fig. 1). In addition, ten oriented samples from two sites in the Bilete complex (Sites 31, 32) were provided by Dr Jean-Paul Karche. Cores, 2.5 cm in diameter were drilled from these and oriented in the laboratory.

All measurements were made on a Schonstedt DSM spinner magnetometer, interfaced to an Apple IIE computer. Demagnetization was performed (a) on a standard two-axis tumbler, with maximum alternating field of 120 mT, enclosed in Helmholtz coils; (b) a Schonstedt thermal demagnetizer with mu-metal shielding. This equipment is now housed (and most measurements were made), in a doubled-walled magnetically shielded (< 1500 gamma) room.

Stepwise AF demagnetization to maximum fields varying from 70 mT to 120 mT was performed on specimens from all samples. Thermal demagnetization was performed on at least one specimen from most sites, and on complete sets of specimens from thirteen sites. These data were analysed using orthogonal projections, and principal component analysis (Kirschvink 1980). The predominant linear regression vectors accepted (Table 1) were those based on no less than four demagnetization steps over a minimum range of 20 mT between 10 mT and 120 mT, or at least four temperature steps between 200° and 680°, with mean angular deviation (MAD) angles of $<10^{\circ}$. Linear segments common to three or more samples from each site were averaged using both Fisher and Bingham statistics (Onstott 1980).

Polished thin sections of at least one sample from each site were studied by transmitted and reflected light to establish their petrological character and, as far as possible, to determine the ferrimagnetic phases which they contain. Study of the opaque phases was facilitated by the use of magnetic colloid.

Results

Palaeomagnetic and petrographic results from each intrusive complex (Fig. 1), summarized in Table 1 and Figs 2–7, are discussed below. The geologic data were extracted from Black, Jaujou & Pellaton (1967) and Husch & Moreau (1982).

1 ADRAR BOUS

At least some samples from all sites in this complex, which consists of both basic and acid igneous rocks, carried anomalously strong NRM's ($>10^{-1}$ emu $cc^{-1} = 10^{-4}$ A m^{-1}) believed to be caused by lightning. Usually, the NRM of these samples could not be recovered by demagnetization, but four out of the six samples from site 22 (troctolite), all showing lightning overprints, gave consistent south-seeking positive vectors, as did samples from another site in this troctolite layer (Site 7; see Fig. 2a). Although fresh, coarsely intergrown ilmeno-magnetite grains are present in these rocks, the stable remanence is considered to be carried by fine-grained magnetite present in serpentinized olivine, and as rod-shaped particles in subsidiary orthopyroxene.

Table 1. Palaeomagnetic results from the Air, Niger.

Complex	Site	N/n	Site Lat N.	Site Long E.	Petrology	J_n	MDF	Linear Vectors D	I	k	α_{95}	North Pole Lat	Long
Adrar Bous	6	6/4	20.3	9.0	Gabbro	(1.6)L	10	6.0	-22.8	45	13.8	58.1	169.3
	6th	6/5											
	7	5/4	20.4	9.0	Troctolite	(31.3)L	10	169.2	+36.0	18	22.4	-48.4	24.4
	7th	5						scattered					
	8	5	20.3	8.95	Adamellite	(5.6)L	<5	scattered					
	21	6	20.3	8.95	Micromonzonite	L	12.5	scattered					
	22	6/4	20.4	9.0	Troctolite	(110)L	7.5	184.5	+49.1	138	7.8	-39.5	4.0
	23	5/4	20.3	8.9	Adamellite	47	5-10	346.4	-32.7	19	21.7	49.6	-150.8
	24	5/4			Gabbro	L	5-12.5	96.7	+66.9	95	9.5	71.9	166.9
	MEAN, SITES 6, 7, 22, 23							356.4	-35.5	44	13.9	49.4	194.2
Tchinidene	43	5	19.2	8.9	Gabbroic anorthosite	18	20	11.9	-55.5	75	8.9	33.6	177.3
	44	5	19.2	8.9	Qtz-feld porph. dike			scattered					
	45	4	19.2	8.9	Basalt dike	5.0E-2	40	03.8	-57.5	128	8.2	32.6	185.4
	MEAN, SITES 43, 45							8.0	-56.6	285	14.9	33.2	181.4
Taguei	3	5	18.8	9.0	Gabbroic anorthosite	(2.3E-1)L	10	scattered					
	25	5	18.8	9.0	Gabbroic anorthosite	(40)L	10	scattered					
Ofoud	4	4	18.75	8.8	Anorthosite	4.0E-1	50	352.7	-47.1	41	14.6	42.4	-162.5
	10	5/4	18.8	8.65	Gabbroic anorthosite	1.4E-1	5-10	04.1	-27.5	47	13.5	56.4	-178.4
	10th	5/4						06.5	-41.4	43	14.1	46.9	-179.9
	11	5	18.8	8.65	Gabbroic anorthosite	1.4	17.5	359.0	-39.1	35	13.1	49.1	-169.8
	11th	5						03.1	-37.1	43	11.9	50.4	-175.7
	5,26		18.8	8.8	Xenocrystic basalt dike	8.0	5	321.6	+12.7	22	14.7	50.6	-94.4
	46		18.8	8.8	Microsyenite dike	3.0	60	scattered					
	46th	4						scattered					
	47	5/4	18.9	8.8	Alkalic granite	1.8E-2	10	350.1	-30.0	55	12.5	53.8	-155.1
	47th	5/3						353.9	-42.6	57	16.4		
	48	5	18.85	8.8	Basic dike	6.0E-2	~10	355.4	+7.0	51	10.9	74.1	-154.2
	48th	5						5.6	-9.1	15	20.7		
	49	6/5	18.9	8.85	Alkalic granite	1.2E-1	<10	358.5	+26.1	199	5.4	84.8	-155.0
	49th	6						05.9	+20.7	18	16.3		
	MEAN, SITES 4, 10, 11, 47							359.3	-42.2	137	7.9	46.7	189.9
Meg.Meg.	12	6	18.6	8.5	Troctolite, olivine gabbro	1.6	30	352.1	-51.6	161	5.3	38.6	-162.9
	50	5/4	18.7	9.0	Basaltic dike	4.0	>60	286.3	+65.6	59	12.1	24.5	-36.6
	50th	5						scattered					
Abontorok	13	5	18.4	8.6	Trachyte	2.0E-3	>50	169.1	-28.7	79	8.7	-79.1	83.7
	14	3	18.4	8.6	Mixed, basic	mixed	mixed	scattered					
	15	5/4	18.4	8.6	Gabbroic anorthosite	2.4E-1	12.5	196.3	+76.2	21	20.3	-6.8	1.4
Iahkout	2	6/5	18.1	8.9	Anorthosite	1.2E-1	10	359.1	+42.2	79	8.7	83.7	1.4
Bilete	1,42	10	18.0	8.3	Trachyte/Syenite	2.6E-1	5-10	scattered					
	1,42th	6						scattered					
	31	5	18.0	8.3	Plagioclase porph. basalt	3.8E-1	30	13.4	-34.8	24	15.9	50.6	168.1
	31th	5						16.4	-47.2	68	9.3	41.0	169.1
	32	4	18.0	8.3	Felsic ignimbrite	4.0E-2	<5	scattered					
	MEAN, SITE 31, AF + TH							14.8	-41.0	143	21.1	45.8	168.7
Guissat	41	6/5	17.8	8.25	Microgranite	1.0E-3	>50	354.3	+43.9	33	13.6	80.7	-25.0
El Meki	51	6/5	17.8	8.3	Granite	1.5E-3	>50	16.3	+12.6	8	28.8	70.2	133.3
Taghouaji	52	6/4	17.1	8.5	Rhyolite porphyry	5.0E-3		262.5	+81.2	12	15.7	14.1	-9.1
	52th	6						scattered					
	53	5/4	17.15	8.5	Alkali granite	1.0E-3	15	249.7	+82.9	21	20.3	11.8	-4.9
	MEAN, 52, 53							256.8	+82.1	219	5.3	13.0	353.0
Zarnouski	54	5/3	14.2	9.6	Tuffsite	7.0	~10	15.5	-13.7	18	20.7	63.9	152.6
	54th	5						50.8	-34.6	5	39.4		
	55	5	14.2	9.6	Microgranite	1.0	~15	scattered				-72.1	-34.3
	56	5	14.2	9.6	Microgranite	5.0	~10	192.3	-2.2	8	29.7		
	MEAN, 54, 56							13.9	-5.8	50	36.1	68.0	149.8

Site = Site number; results given are for AF demagnetization unless suffix th follows Site no.; N/n signifies number of samples collected and number of measured specimens used to obtain vector average. If only one number given, N=n. Site N. Latitude and E. Longitude; J_n = average intensity of NRM, A/m; L signifies that some or all samples from Site are judged to have been struck by lightning; M.D.F. = median demagnetization field which reduces J_n by 50%; D = declination, I = inclination; k = Fisher's kappa, α_{95} = half angle of 95% confidence cone. Calculated North Pole latitude and longitude.

A moderately consistent northerly negative inclination vector was isolated from Site 23 (Fig. 2b), in the westernmost crescentic adamellite ring, and from Site 6, a gabbro exposed in the southern part of the complex (Fig. 2c). The NRM at a second gabbro site (24), however, was completely overprinted by lightning induced IRM.

2 TCHINIDENE

Of three sites sampled in this complex, the gabbroic anorthosite (Site 43) and a ferro-hornblende-basaltic dike (Site 45) gave consistent north-seeking, negative inclination vectors stable to above 100 mT, (Fig. 3a, b). At Site 43, homogeneous ilmenite is the predominant Fe-Ti oxide phase with subsidiary, coarsely intergrown ilmeno-magnetite grains. The augite

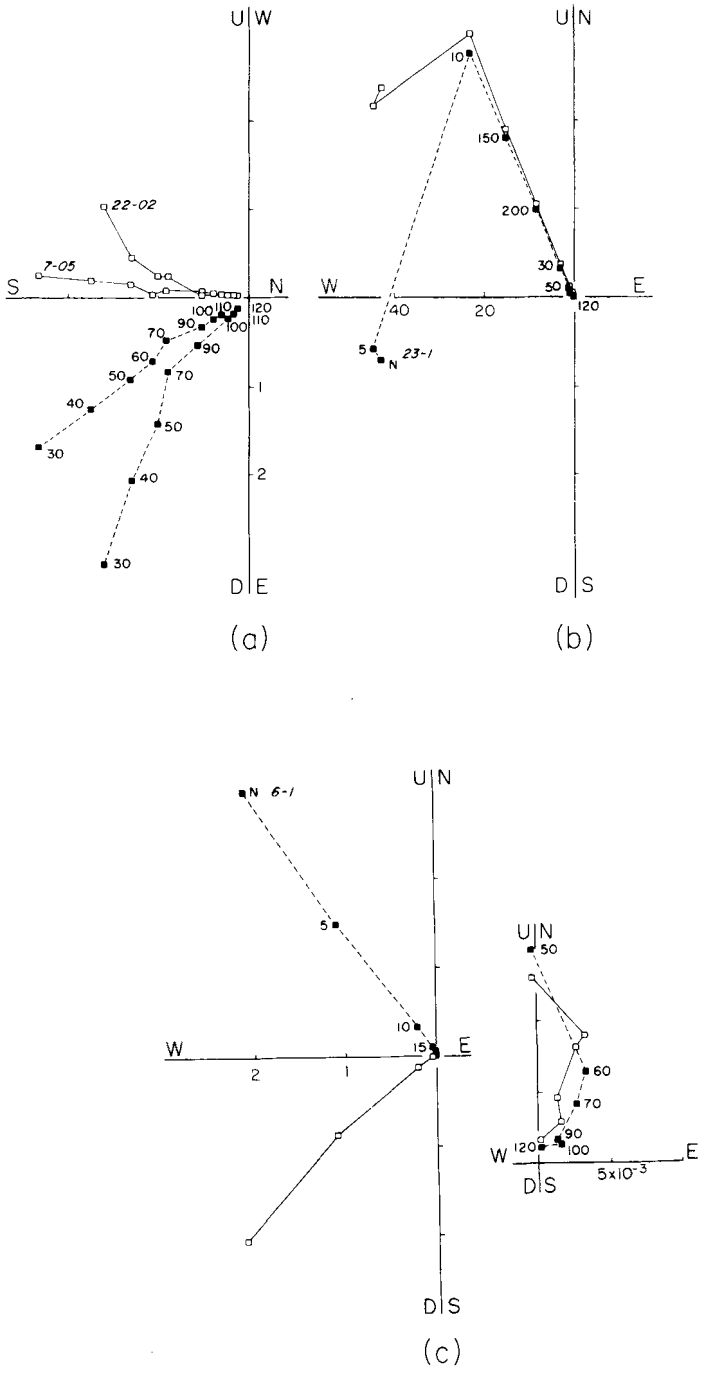


Figure 2. Orthogonal projections of results from AF demagnetization of samples from (a) Sites 22 and 7, (b) Site 23, (c) Site 6, Adrar Bous. Inset to (c) is a projection of results for Site 6 above 50 mT, showing change to northeasterly negative inclination vector; solid/broken lines connecting open/solid circles denote declination and inclination. Intensity units are A m^{-1}.

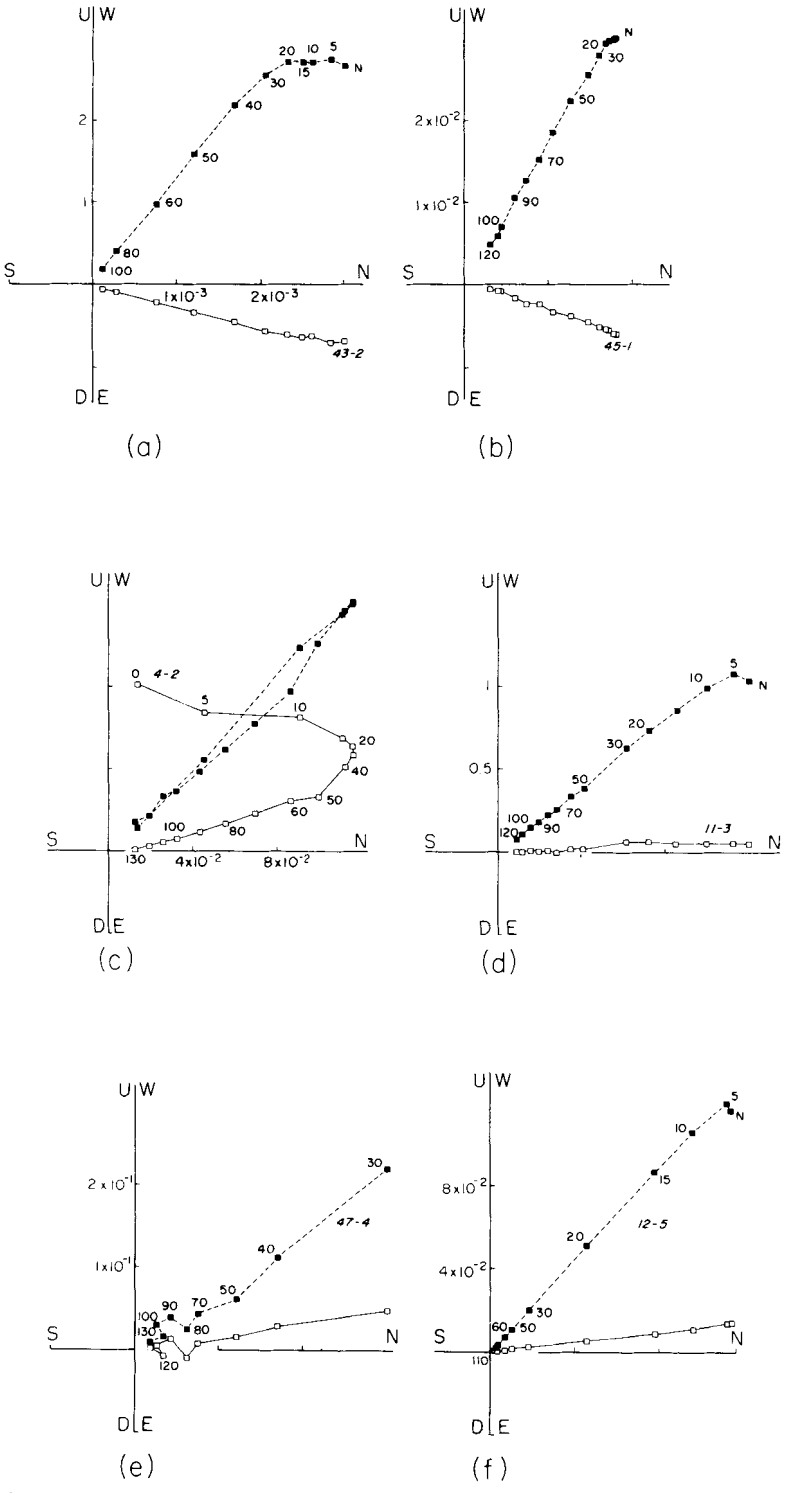


Figure 3. Orthogonal projection of results of AF demagnetization of samples from (a), (b), Sites 43 and 45, Tchinidene, (c), (d), (e) Sites 4, 11 and 47, Ofoud, and (f) Site 12, Meghueur–Meghueur. Symbols as in Fig. 2.

is partly altered to green amphibole, and abundant microscopic opaque inclusions (some of which are magnetic, as evidenced by use of magnetic colloid) are present in both silicate phases. Small grains of pyrrhotite are a minor accessory. The dyke rock of Site 45 consists essentially of green amphibole (with rare augite relics), plagioclase which is intensely altered, about five per cent of homogeneous highly skeletal ilmenite, and minor pyrite. Ilmenite is the only Fe–Ti oxide observed, but rare pyrrhotite grains were detected with magnetic colloid. In addition, there may be fine ferrimagnetic grains in the clouded plagioclase.

3 TAGUEI

The few fresh outcrops at this small anorthosite body (Husch & Moreau 1982) proved to have been completely overprinted by lightning-induced IRM and no useful data were obtained.

4 OFOUD

This is the largest individual ring complex sampled, and contains the greatest exposure of basic rocks (Fig. 1). Three sites in anorthosite (4, Tarenat, 10 and 11, Kori Marit), from opposite sides of the complex, carried internally consistent northerly negative inclination vectors, stable to or above 70 mT AF demagnetization. In addition, samples from sites 10 and 11 have been studied in detail by thermal demagnetization; it is from these data that the most conclusive evidence of the primary TRM nature of the northerly negative magnetization was obtained. These results are summarized below in Tables 2 and 3, and illustrated in Figs 3 and 4.

A soft, low blocking temperature ($<550^{\circ}C$) component is present in all samples. After its removal, a northerly negative inclination component persists to very high alternating fields in samples from Site 4 (Fig 3c, Note: thermal demagnetization was not performed on Site 4 samples). A similar vector persists up to at least 70 mT at Site 11 (Fig. 3d), but thereafter some samples migrate towards a northerly positive inclination direction. Thermal demagnetization of Site 11 samples removes the northerly negative inclination component by $\sim 580^{\circ}C$, leaving the northerly positive inclination component which persists to $680^{\circ}C$.

Table 2. Relation between magnetic components and their coercivity/blocking temperature at Ofoud anorthosite sites.

	Site	Coercivity			
A/F		Low	Moderate	High	Very High
	4	N, +ve		N, -ve	N, -ve
	10	N, +ve	S, +ve	N, -ve	N, +ve
	11	N, -ve		N, -ve	N, +ve
		Blocking Temperature			
Thermal		Low	~Mt	Post Mt	
	10	N, +ve	S, +ve	N, -ve	
	11	N, +ve	N, -ve	N, +ve	

Petrography 4 Ilmenite only, no accessory magnetite observed, but Mt. is present as opaque rods in silicates.

10 Ilmenite > titaniferous magnetite with dense fine trellis ilmenite intergrowth, some grains of which were further oxidized at high temperature to fine rutile + hem ss. intergrowths. Conspicuous opaque rods in cpx; some serpentinization of olivine.

11 Ilmenite > titaniferous magnetite with dense fine (111) trellis intergrowth; fine Mt rods in ilmenite grains closely associated with titaniferous magnetite; Patches, and (111) lamellae of hematite present in the Ti-Mt grains.

Table 3. Detailed thermal and AF demagnetization of two samples from Site 10.

		Steps	D	I	MAD	α_{95}
AF, Sample 1						
10-1.21						
1-3	0-5 mT	4	334.5	51.2	3.5	5.1
4-6	7.5-12.5	3	155.0	43.6	6.0	12.0
7-16	15-60	10	007.8	-42.8	3.9	3.5
17-20	70-11.5	4	195.6	8.4	6.6	18.8
Thermal, Sample 1						
10-1-2						
	25-550°	6	353.6	33.4	4.3	4.7
	555-570°	4	159.3	41.9	15.9	27.4
	575-680°	11	359.5	-35.4	2.2	9.8
AF, Sample 2						
10-2-3						
	0-5 mT	3	344.3	33.4	6.4	5.7
	7.5-17.5	5	143.0	24.5	1.3	4.9
	20-70	9	007.2	-40.9	7.1	13.0
	80-11.5	3	156.3	29.6	14.7	34.9
Thermal, Sample 2						
10-2-2						
	25-550°	6	334.5	39.5	2.4	3.3
	555-570°	4	157.8	34.6	17	7.3
	575-680	11	354.3	-38.4	4.0	20.8

Steps - number of sequential measurements in linear segment. D = Declination, I = Inclination, MAD = Mean angular deviation of points constituting linear segment (Kirschvink, 1980), α_{95} (Kent, Briden and Mardia, 1983).

At Site 10, three components were isolated in individual samples by thermal, and four by AF, demagnetization (see Table 3, Fig. 4a to f). (1) A soft northerly positive inclination vector was removed first by either 5 mT AF, or 550°C thermal demagnetization. (2) A southerly positive component with a very restricted coercivity (7.5–15.0 mT) and blocking temperature (555–575°C) range. (3) A northerly negative inclination vector dominated between 17.5 mT and 60–70 mT, and persisted to 680°C on thermal demagnetization. (4) A southerly positive inclination component reappeared above 70 mT AF demagnetization (Fig. 4e).

The simplest interpretation of these observations is that the very fine-grained magnetite in silicates at Site 4, as well as the small relict magnetite volumes isolated by the dense trellis intergrowth in the high-temperature ilmeno-magnetite and the magnetite rods in the magneto-ilmenite grains at Site 11, acquired a northerly negative-inclination TRM parallel to the ambient field upon cooling. The northerly positive component persisting above 590°C at Site 11 presumably is carried by the low temperature haematite lamellae in the magnetite, and/or the botryoidal haematite overgrowths.

At Site 10, on the other hand, the extreme oxidation of ilmeno-magnetite to intergrowths of rutile and titanhaematite solid solution also occurred at high temperatures ($\geq 600°$) (Haggerty 1976), and therefore this *titanhaematite* acquired a northerly negative inclination TRM upon cooling. Explanation of the two reversed, southerly positive inclination components in Site 10, is less clear. The reversed component with the low coercivity (7.5–15.0 mT) and restricted blocking temperature (555–570°C) is presumably carried by relict primary, multidomain Ti-magnetite, which survived the high temperature oxidation. The other reversed component with very high coercivity (>80 mT) may be carried by haematite or Ti-haematite, the origin of which is obscure. If it is not indicative of field reversal, the southerly positive inclination remanence could be a self reversed component induced in the relict magnetite by magnetostatic interaction with the associated titanhaematite solid solution produced by the high temperature oxidation.

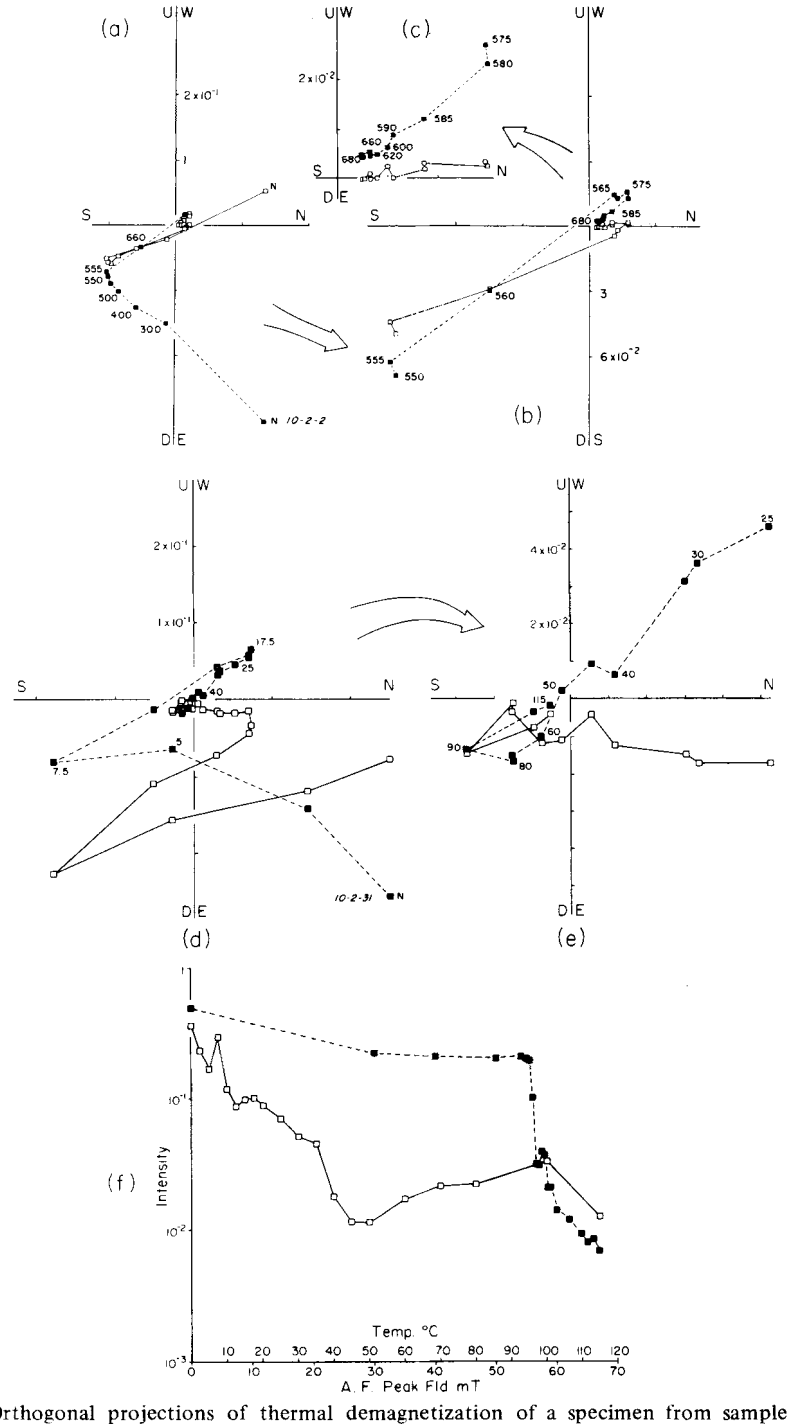


Figure 4. Orthogonal projections of thermal demagnetization of a specimen from sample 2, Site 10, Ofoud. Results from NRM in (a), above 550°C in (b) and above 575°C in (c). Note three distinct components, two of which are nearly antiparallel. (d), (e) show orthogonal projection of AF demagnetization of another specimen from Sample 2, Site 10, Ofoud. Results from NRM in (d) and above 25 mT in (e). Four components are present, two of which are parallel; but separated by removal of a third which is antiparallel. See Table 3. In (f) the variation in intensity (A m^{-1}) on AF (open squares, solid lines) and thermal (solid squares, broken lines) demagnetization are compared. Note oscillations in intensity as opposed components are selectively removed.

The sources of the multicomponent magnetizations present in these Ofoud rocks are seldom easy to identify unambiguously. The abundance or degree of development of the various primary, and high- or low-temperature secondary phases described here, and concomitantly, the coercivity spectra and blocking temperatures of the magnetic components they carry, may vary from sample to sample at individual sites, and even from grain to grain in individual samples. Nevertheless the components can be (and have been) isolated in most cases. Based on magnetic properties and petrographic examination, we believe our identification of those components which were acquired penecontemporaneously with cooling of the rocks (and hence reflect the ambient field at that time), as opposed to those acquired later, is reasonable.

Of the other sites sampled at Ofoud, one (Site 47) in an alkali granite carried a stable northerly negative-inclination vector (Fig. 3e), whereas in another, more weakly magnetized (Site 49), only a northerly positive inclination vector could be isolated. A microsyenite dike (Site 46) proved weak ($\sim 2 \times 10^{-6}$ emu cc^{-1}) and unstable.

Two xenocryst- and xenolith-bearing basic dykes (Sites 5, 26) cutting the Ofoud complex carried only the 'secondary' vector. Whether these are younger (Quarternary?) dykes, or completely overprinted older dykes, is not known.

5 MEGHUEUR–MEGHUEUR

This ~200 m wide ring-dyke, with a diameter of 64 km visible on satellite photographs, is probably the largest in the world. It was sampled at two sites. Site 12, on the western perimeter, included basic xenoliths (dunites) as well as the olivine (micro-) gabbro of the dyke itself, and gave excellent and highly consistent data (Fig. 3f). Because of logistical difficulties, Site 50, an exposure on the east side, was sampled hurriedly, field relations were not clearly established, and Sample 1 was lost. Of the remainder, Sample 2 gave a moderately stable northerly negative inclination vector, but Samples 3–6 carried the same, stable (with respect to AF demagnetization) but anomalous (westerly, and steep positive) vector (Table 1). Detailed thermal demagnetization of these revealed evidence of three distinct components: northerly shallow positive inclination, northwesterly positive inclination and easterly negative inclination, which could not be resolved, and no consistent result was obtained. Whereas Sample 2 is a microtroctolite, the others are olivine dolerite. It is indeed possible that there may have been multiple igneous injections, at different times, into this large ring-dyke, and the four anomalous samples from Site 50 may reflect a separate igneous event.

6 ABONTOROK

This funnel shaped anorthositic intrusion, two and a half kilometres in diameter (Husch & Moreau 1982; Moreau 1982), has a central trachytic core (Site 12) which bears a consistent stable, weak, southerly, shallow negative inclination vector, which is carried by haematite. Site 15 represents a series of samples collected successively inwards from the gabbroic border zone of the intrusion; these gave only moderately consistent vectors, which averaged to very steep southerly (see Table 1). It is not clear that consistent single components have been isolated. Unfortunately, the hand samples from Site 15 have been lost, and all available core material was used in the AF, and combined AF + thermal demagnetization studies. As a result, no further purely thermal demagnetization studies could be undertaken.

7 ISKOUT

A moderately consistent northerly steep positive inclination vector was isolated after analysis of detailed AF demagnetization of the one anorthosite site. As Iskout is within the zone of the Quaternary volcanic activity centred around Todgha (Cantagrel & Karche 1983; see Fig. 1), this vector could represent a stable, but erroneous, hybrid.

8 BILETE

Felsic samples (Sites 1 and 42) gave mixed results. Of the two sites sampled by Dr Jean-Paul Karche, the first (Site 31), a basal plagioclase-phyric basalt flow erupted early in the volcanic cycle, gave a stable (with respect to AF) and consistent northerly negative inclination vector (Fig. 5), whereas the second site was unstable.

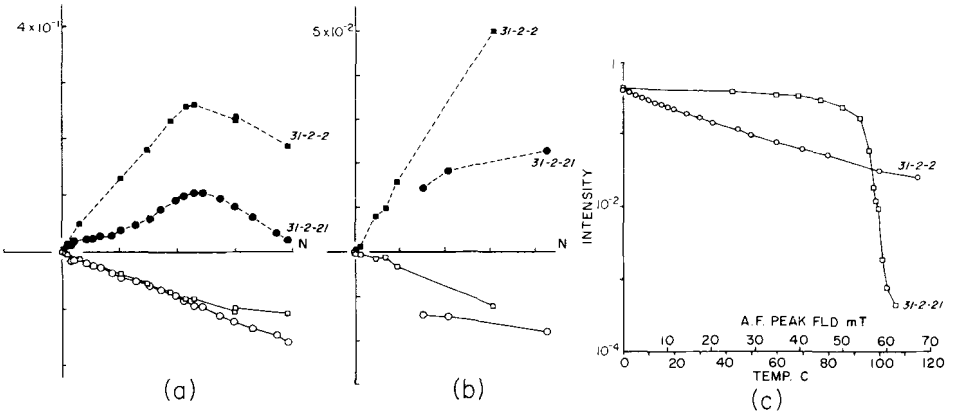


Figure 5. Orthogonal projections of thermal (squares) and AF (circles) demagnetization of two specimens from one samples from Site 31. (a) is from NRM to 620°C, and 115 mT; (b) Starts at 560° and 80 mT. (c) Compares decrease in intensity with thermal and AF demagnetization.

9 EL MEKI AND GUISSAT

These sites, in contiguous rhyolitic or granitic complexes are weak ($\sim 1 \times 10^{-6}$ emu cc^{-1}), but highly stable with respect to AF demagnetization. The El Meki site yielded a moderately consistent, due north and positive vector. The steep inclination obtained from Guissat is almost identical to that recorded from Iskout.

10 TAGHOUAJI

The two sites from this southernmost of the Air complexes gave moderately scattered results both of which averaged, however, to a very steep westerly positive inclination vector. Thermal demagnetization of site 52 gave mixed results, and the presence of two incompletely resolved haematitic components with overlapping coercivities and blocking temperatures, is suspected.

11 ZARNOUSKI

The tiffisite and microgranites (Sites 54, 55 and 56) sampled in the southernmost (and younger, ~320 Myr old) group of complexes in Niger yielded scattered results and no clear or consistent vector could be isolated.

Summary of palaeomagnetic results

The consistent site vectors isolated from the rocks sampled in this study are mostly oriented either northerly, negative inclination (Group A), or northerly, positive inclination (Group B), with a few sites (or samples) giving the reversed equivalent of these. Several other sites give completely distinct vectors, but in these it is not clear that single vectors have been isolated. The poles calculated from each site mean vector are shown in Fig. 6(a).

In general, the strongest NRMs (1×10^{-4} to 1×10^{-2} emu cc^{-1}) are associated with anorthositic, basic-suite rocks, which usually give Group A vectors. Petrologically, these are mostly fresh, with only local and minor alteration of the feldspars and ferromagnesian minerals. Discrete grains of homogeneous ilmenite constitute the predominant primary Fe–Ti oxide, with subsidiary titaniferous magnetite showing trellis, or sandwich, intergrowths between magnetite and oxyexsolved ilmenite; these ilmenitic lamellae have in places undergone further high temperature oxidation. Fine, exsolved, opaque (= magnetite?) rodlets are present in the plagioclase feldspars and pyroxenes of some samples; dusty magnetite (?) has formed in those ferromagnesian minerals which have been serpentinized. Although thermal demagnetization reveals that the stable NRM in these rocks is carried predominantly by nearly pure magnetite, unblocked by 590°C, this vector sometimes persists to temperatures well above 600°C. At Site 10, it is only revealed above 570°C, and in these samples, the ilmenite shows extensive high temperature oxidation to cryptocrystalline intergrowths containing rutile and (titano) haematite, which are presumed also to have formed by oxidation at high temperature (Haggerty 1976). In other Group A samples (e.g. Site 11), the vector remaining above 590°C correlates with the Group B vector. Overall, however, there is no petrographic evidence to suggest that the Group A vector is not a primary TRM or TCRM.

The Group B vector, most common in the acidic rocks, is similar in orientation to the present day or recent geomagnetic field, although the polarity at some sites appears to be reversed (e.g. Site 13). Haematite is the predominant carrier of this component, and can be seen in polished section as veins in magnetite (Site 11), or as reddish alteration of ferromagnesian silicates. Although this may in part be due to recent weathering, we consider it more likely to the effects of the extensive Quaternary volcanism centred around Todgha (Cantagrel & Karche 1983; see fig. 1).

The meaning of the other, apparently consistent, anomalous vectors isolated at other sites remains unclear.

Structural control

As these palaeomagnetic data are nearly all (except for Site 31) derived from intrusive igneous rocks, lacking direct evidence of a palaeohorizontal datum, their palaeogeographic significance depends upon the validity of the assumption that they have not undergone tectonic rotation since they acquired their NRM upon cooling. The absence of such a datum in plutonic igneous or high-grade metamorphic rocks is commonly judged to vitiate their value in palaeomagnetic studies. We believe, however, that there are distinct advantages in the palaeomagnetic study of such rocks. While the absence of a direct palaeohorizontal indicator is acknowledged, professional judgement may be made as to the likelihood that significant cryptic tectonic rotations have or have not occurred. The best argument in favour of the latter is demonstrating that the vectors recovered from penecontemporaneous plutonic intrusions, distributed over as wide an area as possible, are essentially the same. This is not proof of the absence of tectonic rotation, but it is a reassuring indication.

The overwhelming advantage of the palaeomagnetic measurement of plutonic rocks derives from their relatively coarse grain. This facilitates petrographic study of the ferrimagnetic phases present within them, and also the separation of those mineral phases which may be independently dated. In contrast, the direct study of ferrimagnetic phases in supracrustal rocks is difficult, and the dating of the magnetization they may carry is fraught with uncertainty.

In this study a similar, apparently primary, bipolar Group A vector has been recovered (Table 1) from twelve sites distributed over ~300 km. The slight differences between the average vector from different complexes (Fig. 6b) may reflect minor tectonic rotations, or temporal differences. For example, the basal lava flow (or sill?) sampled at Bilete dips ~15° to the east (Karche, personal communication), but its northerly (Group A) vector is virtually unaffected by the structural correction. These observations reassure us that significant tectonic rotations, which might have seriously compromised the palaeomagnetic study, have not occurred.

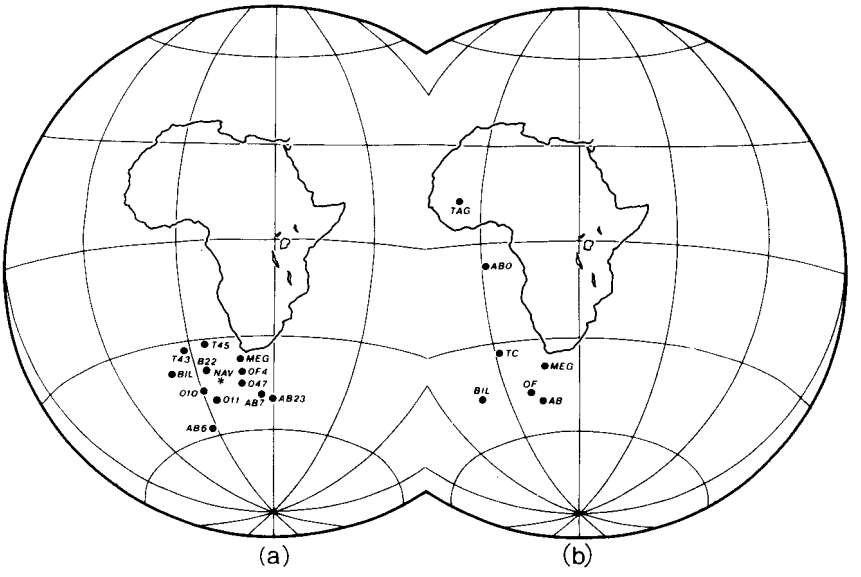


Figure 6. (a) Individual site poles (n = 12) used to calculate Niger mean pole (= NAV), shown by star. Adrar Bous (AB or B) sites 6, 7, 22 and 23; Tchinidene (T) sites 43 and 45; Ofoud (O or OF) sites 4, 10, 11, 47; Meghueur–Meghueur (MEG) Site 12; and Bilete (BIL) Site 31. (b) Group mean poles (Table 1): AB = Adrar Bous, TC = Tchinidene, OF = Ofoud, MEG = Meghueur–Meghueur, ABO = Abontorok, TAG = Taghouaji.

Age of the group A magnetization

The published radiometric data from Niger pertinent to this study (all obtained from granitic rocks) are summarized in Fig. 7. These have been interpreted in terms of igneous activity starting at 480 Myr in the north and migrating steadily southward (Karche & Vachette 1978; Bowden & Kinnaird 1984). The concentration of ages around 480 Myr in the Air (Fig. 7) should, however, be noted. The similar polarity at ten of the twelve sites carrying Group A magnetization and the similarity of all pole positions favours penecontemporaneity of magnetization. From independent palaeomagnetic and geologic evidence, the south pole

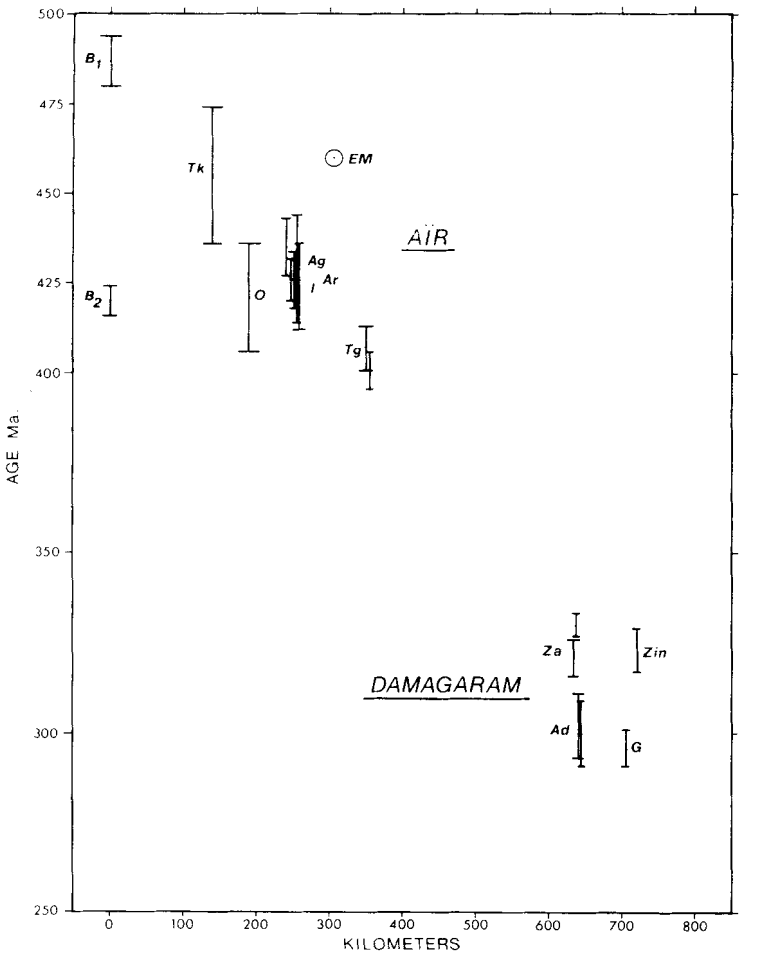


Figure 7. Radiometric ages (Rb–Sr whole rock from Karche & Vachette 1978; line length signifies reported error) from Niger ring complexes plotted as a function of distance to the south from Adrar Bous in the north (Fig. 1). B = Adrar Bous; Tk = Tamgak; M.M. = Meghueur–Meghueur; O = Ofoud; AG = Agalak, Ar = Aroyan, I = Ishkout; EM = El Meki (K–Ar, Brunnschweiler 1974), Tg = Taghouaji; Za = Zarnouski; Zin = Zinder; G = Goure; Ad = Adoumchi.

with respect to Gondwana was located in North Africa (if not further north) in early Ordovician time (Livermore, Smith & Briden 1985; Scotese, Van der Voo & Barrett 1985); this precludes correlation of the NRM at Adrar Bous, the northernmost complex, with the 480 Myr age reported. However, a radiometric age of 480 Myr has also been reported for this complex (Bowden *et al.* 1976) and hence the acid igneous activity, even if it started in the early Ordovician, evidently persisted until mid–late Silurian time.

The overall similarity of the Adrar Bous poles and those obtained farther south gives further support to the ~420 Myr age reported for the former complex, as the same Rb–Sr age has also been obtained in Ofoud (Bowden *et al.* 1976). As is evident from Fig. 7, igneous activity around 420–425 Myr was widespread in the Air. Taghouaji, from which no good palaeomagnetic data were obtained, has been dated at ~ 400 Myr old, and is considerably farther south (Fig. 1). From these published data, the best estimate of the age of the evidently primary TRM (or TCRM) would be ~420 Myr.

Our palaeomagnetic data were obtained from basic igneous rocks, however, and in each complex the emplacement of these is reported to have preceded the intrusion of the alkalic felsic rocks (Black, Jaujou & Pellaton 1967). To determine the age of the magnetization directly, an $^{40}Ar/^{39}Ar$ study has been initiated to date whole rocks or mineral separates from samples used in this palaeomagnetic study. Results obtained on biotite separated from samples of anorthosite and gabbroic anorthosite (Sites 10 and 11, in Ofoud) and a gabbro from the ring dyke Meghueur–Meghueur (Site 12) are presented in Fig. 8 and Table 4. The plateau age for the Ofoud anorthosites are 438.8 ± 4.4 Myr and 436.0 ± 4.3 Myr, whereas that for the ring dyke is slightly younger at 431.1 ± 4.3 Myr. More $^{40}Ar/^{39}Ar$ age measurements of samples from other complexes are in progress, and these data are needed before a conclusive interpretation can be made as to the timespan represented by the palaeomagnetic data. But inasmuch as the closure temperature for diffusion of argon in biotite is generally regarded as ~300°C, these three biotite ages indicate that the rocks have not been heated in excess of this temperature since ~435 Myr which, we conclude, is the age of the Group A magnetization in the Air rocks.

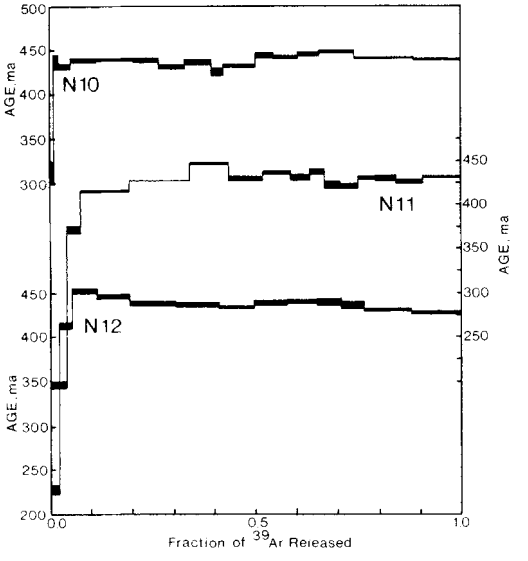


Figure 8. $^{40}Ar/^{39}Ar$ incremental release spectra for biotite separated from gabbroic anorthosite and anorthosite sites N10 and N11 Ofoud, and gabbro from the ring dyke Meghueur–Meghueur, Site N12.

Middle Palaeozoic apparent polar wander path for Gondwana

There is a conspicuous dearth of Silurian–Devonian palaeomagnetic poles from the Gondwana continents (Scotese, Van der Voo & Barrett 1985). What data there are, have been largely obtained from the Lachlan Fold Belt in southeast Australia (Fig. 9, Table 5). As the relationship of this terrane to cratonic Australia prior to the Devonian is not clear, several different APW curves for Gondwana in the Early Palaeozoic have been proposed (see Livermore, Smith & Briden 1985, fig. 2). McElhinny & Embleton (1974) excluded the southeast Australia data, and proposed that the pole migrated slowly down Africa from the north in late Ordovician time to central Africa in the late Devonian; this model has been favoured by most recent workers (Perroud, Van der Voo & Bonhommet 1984; Livermore,

Table 4. $^{40}Ar/^{39}Ar$ analytical data.

T(°C)	f_1	f_2	Atmos (%)	$^{40}Ar^*$ ($\times 10^{-6}$ cm^3 STP)	Cum. f(39)	$^{37}Ar_{Ca}/^{39}Ar$	AGE (Ma)
			NIGER 10 BIOTITE				
550.0	0.00046	0.00207	51.4	0.877	0.010	0.701	312.6±13.1
600.0	0.00023	0.00648	12.2	1.408	0.021	-0.351	436.5±8.4
650.0	0.00009	0.00717	4.6	3.784	0.050	0.132	431.1±2.4
700.0	0.00002	0.00960	1.0	8.443	0.115	0.038	437.6±2.2
750.0	0.00003	0.27506	0.0	11.345	0.201	0.039	438.6±1.3
775.0	0.00003	0.11533	0.1	7.930	0.262	0.045	437.5±1.8
800.0	0.00005	0.08632	0.2	8.267	0.326	0.080	430.6±2.2
825.0	0.00001	0.05794	0.1	8.424	0.391	0.020	437.5±1.6
850.0	0.00006	0.12501	0.2	8.678	0.420	0.095	424.8±4.0
875.0	0.00002	0.00910	0.9	10.013	0.498	0.032	432.2±1.4
900.0	0.00005	0.01346	1.5	5.786	0.541	0.081	444.0±3.1
950.0	0.00004	0.01106	1.6	7.864	0.601	0.072	441.9±2.0
1000.0	-0.00010	0.00981	3.8	6.944	0.653	0.151	445.4±2.2
1050.0	0.00008	0.00893	3.5	11.697	0.740	0.126	448.1±1.6
1100.0	-0.00004	0.00394	4.1	18.873	0.884	0.066	440.6±0.7
1150.0	0.00004	0.00252	5.3	15.259	1.000	0.054	439.2±1.5

Sample Mass = 0.0274 Grams
Average Standard J Value = 0.009654±0.000015
Integrated Date = 437.6±4.4 (Ma)
Plateau Date (600-1150°C) = 438.8±4.4 (Ma)

			NIGER 11 BIOTITE				
550.0	-0.00013	0.00190	41.2	1.440	0.041	0.197	176.7±1.7
625.0	-0.00002	0.00118	7.2	2.546	0.072	0.030	376.7±4.1
700.0	-0.00001	0.00108	2.6	10.803	0.190	0.010	418.4±1.1
750.0	-0.00001	0.00344	1.0	13.883	0.337	0.013	430.7±0.6
775.0	0.00000	0.00104	0.5	9.511	0.433	0.002	449/7±1.1
800.0	0.00000	0.00784	0.2	7.824	0.515	0.007	432.8±1.4
825.0	-0.00002	0.04640	0.2	6.512	0.583	0.033	438.6±1.7
860.0	0.00005	0.01399	1.3	4.476	0.630	0.069	432.3±3.1
900.0	0.00003	-0.00378	2.7	3.431	0.665	-0.040	443.2±2.6
950.0	-0.00009	0.00663	5.4	3.463	0.702	0.140	429.4±3.8
1000.0	-0.00011	0.00573	7.3	4.117	0.746	0.169	428.1±2.2
1040.0	0.00008	0.00367	8.2	4.037	0.789	0.126	438.4±1.4
1075.0	-0.00009	0.00329	9.5	4.911	0.841	0.133	438.8±2.5
1110.0	-0.00009	0.00321	10.6	6.164	0.906	0.145	435.2±1.9
1150.0	0.00020	0.00752	9.6	8.928	1.000	0.307	434.5±0.8

Sample Mass = 0.0429 Grams
Average Standard J Value = 0.009654±0.000015
Integrated Date = 422.2±4.2 (Ma)
Plateau Date (750-1150 °C) = 436.0±4.3 (Ma)

			NIGER 12 BIOTITE				
625.0	0.00008	-0.00227	21.0	1.118	0.020	-0.116	222.7±4.7
700.0	-0.00002	0.00184	4.7	3.502	0.052	0.033	406.2±3.3
750.0	-0.00000	0.00010	0.1	6.744	0.108	0.000	446.7±2.2
775.0	-0.00008	0.47541	0.0	9.660	0.190	0.130	441.8±2.6
800.0	-0.00001	0.00881	0.4	13.011	0.303	0.012	433.3±1.6
820.0	0.00000	-0.00977	0.2	12.138	0.408	-0.006	432.3±2.1
840.0	-0.00002	0.11315	0.1	9.963	0.496	0.026	429.7±1.5
865.0	0.00000	0.05505	0.0	9.035	0.574	-0.006	435.0±2.0
900.0	0.00003	-0.37069	0.0	8.353	0.646	-0.045	434.9±1.9
940.0	0.00005	-0.00680	2.7	6.801	0.704	0.071	434.1±3.0
1000.0	-0.00008	0.00641	4.9	6.339	0.759	0.128	431.5±3.3
1075.0	0.00007	0.00584	4.8	13.083	0.875	0.112	427.1±1.5
1175.0	-0.00006	0.00338	7.1	14.042	1.000	0.097	426.5±1.6

Sample Mass = 0.0358 Grams
Average Standard J Value = 0.009654±0.000015
Integrated Date = 428.1±4.3 (Ma)
Plateau Date (800-1175 °C) = 431.1±4.3 (Ma)

Notes:
f1 = [37/39)Ca/(37/39)M], and f2 = f1·(36/39)Ca/(36/39)M], where
(Ca = isotope ratio of argon extracted from irradiated calcium salts and
M = isotope ratio of argon extracted from irradiated unknown
f(39) = cumulative fraction of 39Ar released in each step.
Date(Ma) = the date calculated using the following constants:
Lambda,epsilon = 0.581E-10/year; lambda/beta = 4.96E-10/year
40K/K = 0.01167 atom % (Steiger and Jager, 1977).
The quoted error is one standard deviation and does not include the error in the J value or the standard error. Integrated Date = date and error calculated from the sum total from all steps; the error includes the error in the J value. Plateau Date = date and error calculated from the sum total from those steps, the dates of which fall within 2 S.D. of each other; the error includes the error in the J value.

Smith & Briden 1985; Scotese, Van der Voo & Barrett 1985). In contrast, Schmidt & Morris (1977) and Morel & Irving (1978) both considered that the southeast Australian data might be acceptable for Gondwana as a whole, Morel & Irving (1978) proposing an alternative (Path Y) in which the pole moved rapidly to the south–southwest of South America during Silurian time, before returning to central Africa in the Late Devonian.



Figure 9. Plot of all reported Silurian–Devonian poles for Gondwana rotated according to fit of Smith & Hallam (1979). Source of data: circles – Africa; squares – South America; upright triangles – Australia; inverted triangles – Antarctica; star – Air (Niger) pole; large upright triangle – Canning Basin pole (Hurley *et al.* 1985).

In a recent paper, however, Schmidt *et al.* (1986) reported a reliable (positive fold test) mid–late Devonian pole from southeast Australia, which, for lack of geological evidence to the contrary, they are confident does apply to cratonic Australia (and hence to Gondwana) at that time. This pole (Comerong Volcanics, CV, Fig. 9) lies somewhat north of our Air pole. Schmidt & Embleton (1985) also report what is apparently an equally dependable (positive fold test, etc.) early Devonian pole from the Snowy River volcanics, in southeast Australia, which is even closer to our Air pole (SR, Fig. 9). These new middle Palaeozoic palaeomagnetic data from southeast Australia clearly support a southerly excursion of the Gondwana APW path if the Lachlan Fold Belt was joined to the Australian craton at the time the rocks were magnetized. Schmidt *et al.* (1986) attribute the contrasting central African location of other Devonian poles from Australia (BG, CB, MS, HG and MD; see Fig. 9) to remagnetization in Carboniferous time. [It should be noted that the Moroccan Msissi Norite pole also located in central Africa – and hitherto thought to be Devonian – is now discounted because the intrusion has been radiometrically dated at ~ 140 Myr and its magnetization varies from one outcrop to another (Salmon *et al.* 1986).]

Our Silurian Air result gives particular support to Morel & Irving's (1978) path Y (or some variant thereof), because it comes from the African craton and does not have the structural uncertainty which might be associated with southeast Australian data. We recognize, however, that its age of ~435 Myr requires exceptionally rapid northward drift of the whole of Gondwana if, as most reconstructions accept, the late Ordovician pole (~450 Myr old?) was in northern Africa. Such a location is accepted on the basis of available Ordovician palaeomagnetic data, and also because extensive late Ordovician glacial deposits are preserved there. However, it should be noted that the Ordovician palaeomagnetic poles

Table 5. Lower–Middle Palaeozoic Gondwana South Poles (rotated where appropriate according to Smith & Hallam 1970).

LAT	LONG	REF	SYMBOL	NAME AND AGE
AFRICA				
47.0	42.0	1	AA	Anti Atlas 532±18 Ma, U/Pb
43.4	08.6	4	AIR	Air intrusives 435 (⁴⁰Ar/³⁹Ar,
51.0	353.0	1	BL	Blauhoker fm 471 Ma
13.6	23.5	3	BO	Bokkeveld O
36.0	52.0		CB	Plateau d'Ouefl Gp. 435 465 Ma(K Ar)
27.5	351.4	2	DAM	Damara belt 450 550 Ma (Rb Sr)
36.7	352.5		GD2	Gannakouriep Dykes 592±4 Ma (AA)
-35.2	43.6	6	GN	Gneiguira redbeds Dm
26.5	10.2	3	GR	Graafwater TM group O
14.0	336.0	1	HK	Hook intrusives 472 Ma (K Ar Bi)
53.0	26.0	1	IBM	Hasi messaud srds O Ca
28.0	345.0	1	NT	Ntonya ring structure 523 536 Ma (K Ar, Hal) >481 Ma(K Ar)
37.1	42.6	5	SAB	Sabaloka 399±7 Ma
SOUTH AMERICA				
17.4	77.6	7	AL	Alraparrosa Ou 416 Ma, 291 Ma(K Ar,W)
1.8	344.1	9	B1	Bolivian seds O
7.3	346.1	9	B2	Bolivian seds Dl(O?)
83.1	4.5	7	CNT	Campanario north Tilcara Ca
85.5	337.6	7	CST	Campanario south Tilcara Ca
26.4	1.6	9	JU	Jujuy redbeds O
22.6	21.4	8	PTQ	Piquete complex 532-468 Ma; Peak 495 Ma (K Ar)
17.9	11.2	9	PP	Pitos Passagem series Dl
24.8	0.5	9	SA	Salta Jujuy redbeds Ca,O
24.9	47.4	7	SU	Suri Ol
39.4	13.2	9	UR	Ururum fm O(S)
AUSTRALIA				
21.1	19.6	10	AB	Arcoma dam seds Cal
-17.4	05.8	10	AV	Ainslie volcanics Dl
25.8	319.3	10	APV	Antrim plat volc Cal
7.5	24.1	10	BG	Bowning Gp Dl
3.4	22.1	12	CR	Canning Basin reef Du
25.8	05.4	11	CV	Comerong volcanics Du
31.5	358.3	10	DG	Dundas Gp Cau
71.5	338.7	10	HF	Hudson Fm Cam
20.5	39.8	10	HG	Housetop granite Dm 375 Ma
67.5	33.1	10	HHS	Hugh river shale Cal m
43.7	09.7	10	JF	Jinduckin Fm Ol
40.8	10.2	10	LFG	Lake Frome gp Cam u
13.2	51.1	10	MD	Mulga Downs Du
26.3	09.3	10	MP1	Mugga porphyry Su 423 Ma
15.7	21.9	10	MS	Mereenie ss S? D
48.5	13.5	11	SR	Snowy river volcanics Dl
54.4	51.7	10	SS	Stairway ss Dm
43.4	333.6	10	SV	Silurian volcanics Sm u
0.8	343.3	10	TRD	Todd river dolomite Cal
27.1	17.8	10	TS	Tumblagooda ss O?(Cam-Sl)
ANTARCTICA				
49.2	15.7	10	MS	Mirnyy station charnockite 520±24 Ma (Rb/Sr)
17.5	15.3	10	SRM	Sor Rondane intrusives >489 Ma(Rb-Sr)
42.2	18.5	10	TV	Taylor valley lamprophyre 470Ma?

References: (1) Broche (1982); (2) Corner (1983); (3) Bachtadse, Van der Voo & Kroner (1984); (4) This manuscript; (5) Saradeth, Schult & Soffel (1985); (6) Kent, Pia & Sougy, 1984; (7) Vilas (1981); (8) Filho, Sato & Pacca (1985), Filho, Pacca & Sato (1986); (9) McElhinny (1973); (10) Embleton (1981); (11) Schmidt & Embleton (1985), Schmidt *et al.* (1986); (12) Hurley & Van der Voo (1985), Hurley *et al.* (1985).

for which there are radiometric dates (Fig. 10, Table 5) were in that region during the *early* part of the period. On this evidence, and to be more compatible with our new early Silurian pole far to the south (Fig. 9), we suggest that the poorly constrained late Ordovician pole was in western central Africa, a position similar to that inferred by Morel & Irving (1978) for their Path Y. According to this hypothesis the Late Ordovician and earliest Silurian glacial deposits in northern, southern, and western central Africa accumulated near the *fringe* of a



Figure 10. Plot of all Cambrian–Ordovician poles for Gondwana, with radiometric ages; K = K–Ar, R = Rb–Sr, A = Ar–Ar, U = U–Pb; Bi = biotite, Hbl = hornblende, WR = whole rock; other symbols as in Fig. 9.

continental glacier. By analogy with the Pleistocene record, such deposits could have formed as much as 40° from the earth's rotational pole (Van Houten & Hargraves 1987).

Palaeogeography and tectonics during Early and Middle Palaeozoic time

Two quite different, yet remotely related events occurred synchronously during Mid-Palaeozoic time. One was an orogeny[1] in eastern Laurentia (North America) and southern Baltica (northwestern Europe); the other, a congenial climatic episode in western Gondwana. Reconstruction of the first activity depends in part, at least, on knowing the time and mode of emplacement of Armorican blocks[2] along the margin of Laurentia and of Baltica. The

[1] The Acadian Orogeny in eastern Laurentia and the late phase of the Caledonian Orogeny in southern Baltica occurred during early to middle Devonian time (Dallmeyer & Gee 1986; Kent 1985; Perroud, Van der Voo & Bonhommet 1984; Schenk & Lang 1982; Ziegler 1978).

[2] Armorican blocks (Scotese *et al.* 1979; Van der Voo 1982) include in present day Europe: Italy, Bohemia, Thuringia, and southwestern Poland, northwestern Spain and adjacent Portugal, Normandy and Brittany (whence the name Armorica), southern England and adjacent Belgium, northern Wales and southern Ireland; in present day North America: eastern Newfoundland and Grand Banks, northern and western Nova Scotia, eastern New Brunswick and New England, and the Piedmont and Florida; and in Africa: the Moroccan mesetas. The North American blocks included here have been named Avalonia or Avaland after the Avalon Peninsula of eastern Newfoundland. Armorican blocks in Europe and Morocco were involved in Late Palaeozoic orogeny and are commonly referred to as Variscan mesetas or Hercynian massifs. Palaeontologic (Cocks & Fortey 1982; Neuman 1984) and palaeomagnetic (Perroud, Van der Voo & Bonhommet 1984) data suggest that all of these Armorican blocks were in high southern latitudes in Early Palaeozoic time, and were separated from Baltica and Laurentia.

second event intervened between widespread early and limited Middle Palaeozoic glaciation. Attempts to understand these events have been hampered largely by uncertainty about the position of Gondwana at that time (Scotese, Van der Voo & Barrett 1985).

A lack of constraining data has led to several contrasting Palaeozoic reconstructions. In one, which is indifferent to palaeomagnetic data, Pangaea was an assembled super-continent throughout the Era, with the Armorican blocks already contiguous to Laurentia, Baltica and Western Gondwana in Mid-Palaeozoic time (Boucot & Gray 1980; Boucot 1985). In a second scenario, which relies heavily on palaeomagnetism, Western Gondwana, the Armorican blocks, and Laurentia–Baltica (Laurussia after early Caledonian closure) were not assembled until either Middle or Late Palaeozoic time. Among reconstructions of this latter sort, in one the Armorican blocks alone drifted northward and collided with Laurentia and Baltica by middle Devonian time, leaving Western Gondwana in high southern latitudes, to converge with them only later in the Carboniferous Period (Perroud, Van der Voo & Barrett 1984). In a second, the Armorican blocks and Western Gondwana together converged with Laurussia in Mid-Palaeozoic time to form a Pangaea. Two variations of this are: (a) Western Gondwana remained there until Early Mesozoic time (see Livermore, Smith & Briden 1985, figs 6, 7; Scotese, Van der Voo & Barrett 1985, figs 3, 4), or (b) it drifted southward again during late Devonian time, opening an ocean; this, in turn, was closed by northward drift of Western Gondwana during the Carboniferous Period (Morel & Irving 1978). In yet a third variant, Western Gondwana *and* the major Armorican blocks were still separated from Laurussia by an ocean in Mid-Palaeozoic time, then drifted northward to assemble Pangaea in latest Devonian time (see Livermore, Smith & Briden 1985, figs. 5, 7).

Our new, well-documented early Silurian (~ 435 Myr) pole may help to resolve these contradictory histories.

On the basis of the information now available, and recalling the longitudinal indeterminacy of reconstructions based on palaeomagnetic poles, we suggest the following succession of events as Western Gondwana moved toward and across the South Pole during the Palaeozoic Era (Figs 11, 12; Van Houten & Hargraves 1987). In Cambrian time the widely separated continents of Laurentia, Baltica, and Western Gondwana with contiguous Armorican blocks, lay in low latitudes (Morel & Irving 1978; Smith, Briden & Hurley 1981).

During Ordovician time northern Africa drifted toward the South Pole so that all of Western Gondwana and the Armorican blocks were in a cold-water realm (Boucot & Gray 1980). Toward the end of the Period the pole lay far inland (Fig. 11A), and widespread continental glaciation in Western Gondwana (Caputo & Crowell 1985; Cocks & Fortey 1986; Hambrey 1985) was accompanied by a global low stand of sea-level (Vail, Mitchum & Thompson 1977; Hallam 1984).

During latest Ordovician and early Silurian time Africa continued drifting rapidly across the pole while glacial deposits accumulated farther southwest in Western Gondwana (Caputo & Crowell 1985). By early Silurian time the South Pole was located at the southern margin of South America (the Air position), the Armorican blocks lay in low latitudes, but Gondwana was located east of Laurussia (Fig. 11B). Glaciation ended altogether during early Silurian time as the pole approached the 'southern' coast of Gondwana; a warmer water realm encompassed the northern part of the region (Fig. 11B; Boucot & Gray 1980). This episode of rapid northward drift, together with melting of the ice cap, led to a global *high* stand of sea level (Fig. 12; Vail, Mitchum & Thompson 1977; Hallam 1984; Boucot 1985).

We propose that the pole remained approximately stationary with respect to Gondwana throughout the Silurian Period. If the southeast Australian Snowy River pole (390 Ma) does apply to Gondwana as a whole at this time, then its near coincidence with our Air pole (see Fig. 9) supports this hypothesis. This does not mean, however, that Gondwana was

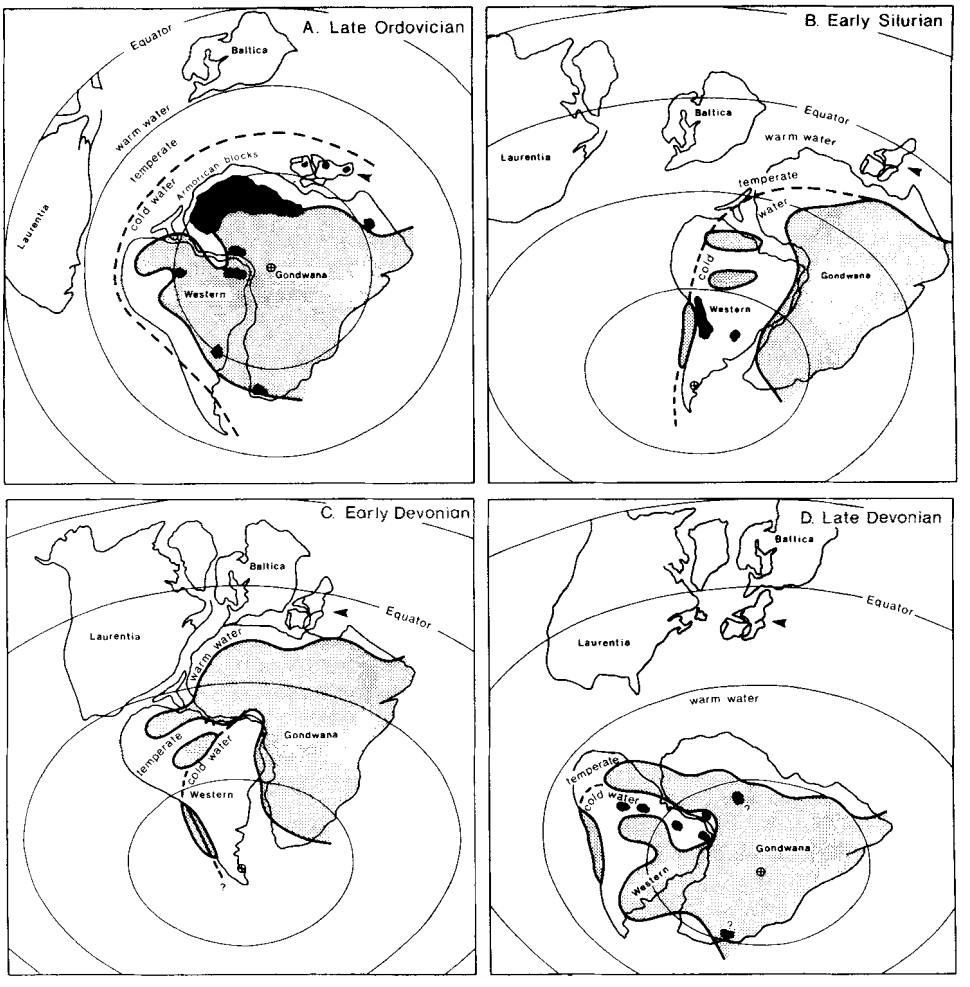


Figure 11. Palaeocontinental reconstructions showing changing relations of Laurentia, Baltica, Western Gondwana (rest of Gondwana omitted for convenience of plotting), and Armorican blocks (largest, central and southern Europe, indicated by arrowhead). Major land area of Western Gondwana is stippled. Black patches are glacial deposits. Except as noted north pole positions were derived from Livermore *et al.* (1985, table 3), using the following rotations:

	lat.	long.	angle
EUR–AFR Pangaea A-2	41.4	354.3	59.3
NAM–AFR Pangaea A-2	57.8	336.5	89.1

(A) Late Ordovician Period. Pole for Laurentia (Scotese *et al.* 1979): 27N, 112E; for Baltica: 10.3N, 187.1E; for Western Gondwana (our estimate, see text): 5S, 195E. Ocean temperature realms mainly after Boucot & Gray (1980, fig. 2). Glacial deposits are Late Ordovician and earliest Silurian. (B) Early Silurian Period. Pole for Laurentia (interpolated from Livermore *et al.* 1985): 30.7N, 112.2E; for Baltica (interpolated from Livermore *et al.* 1985): 16.9N, 173.7E; for Western Gondana (Air, Niger, pole): 43.4N, 188.6E. Ocean temperature realms estimated from Boucot & Gray (1980, fig. 2 and 3) and Boucot (1985, fig. 2). (C) Early Devonian Period. Pole for Laurentia: 30.2N, 99.4E; for Baltica: 23.5N, 160.4E; for Western Gondwana (Air, Niger, pole): 43.4N, 188.6E. Ocean temperature realms estimated from Boucot (1985, figs 1, 2) and Johnson (1979, fig. 6). (D) Late Devonian Period. Pole for Laurussia: 35N, 162E; for Western Gondwana (Hurley *et al.* 1985): 3.4S, 202.1E. Ocean temperature realms as in late Middle Devonian after Johnson (1979, fig. 7).

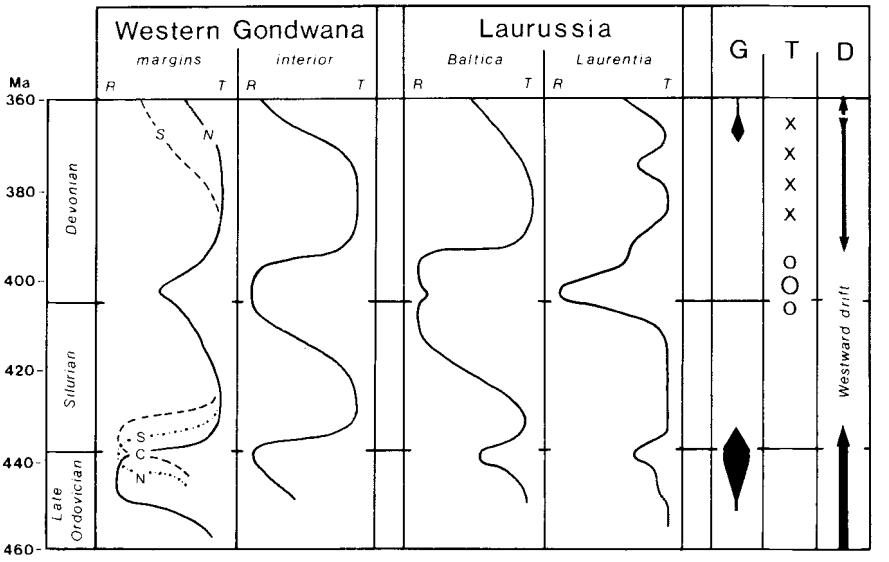


Figure 12. Major Middle Palaeozoic events. Adapted in part from Boucot (1985), Hallam (1984), and Palmer (1983). R – regression, T – transgression; N – north, C – central, S – southern; G – Gondwanan glaciation; T – tectonic activity; X – extensional phase along southern margin of Laurussia and northern margin of Western Gondwana, O – late Caledonian orogeny; D – rate and north and south drift of western Gondwana, width of arrow is proportional to estimated drift rate.

immobile. Instead, we believe that throughout this interval the Euler rotation pole for the 'Gondwana' plate coincided approximately with the geographic pole. Gondwana rotated westwards (relatively) about this Euler pole to converge on and collide with Laurussia in the late Silurian–early Devonian (Fig. 11C). During this late Caledonian Orogeny, continental margins experienced extensive regression (Fig. 12).

Later in the Devonian Period Gondwana moved southward again (Fig. 11D) leaving behind some of the Armorican blocks attached to Laurussia. This separation coincided with an episode of transgression, and of extension along the continental margins of southern Laurussia and northwestern Africa (Fig. 12) (Leeder 1976; Wendt 1985; Ziegler 1979). A temperate carbonate facies near the southern limit of shelf carbonate accumulation in northwestern Africa (Wendt 1985) gave way to an even cooler deltaic to shallow marine detrital facies by latest Devonian time. When central Africa arrived at the South Pole in late Devonian time the cold-water realm reached northern Brazil (Johnson 1978) and at least local glaciation spread from northern Brazil (Caputo 1985) southeastward across Africa during a major regression (Fig. 12). In early Carboniferous time Gondwana began to move northward again, and as the pole shifted southeastward, glaciation became more widespread, leaving deposits in northern Brazil, western central South America, and central and southeastern Africa (Hambrey & Harland 1981; Caputo & Crowell 1985).

We submit that this scenario is consistent with both the stratigraphic and palaeomagnetic record, and it accounts for significant early and middle Palaeozoic events in Western Gondwana and Laurussia. Moreover, it includes a mid-Palaeozoic Pangaeic assemblage like that proposed by McKerrow & Ziegler (1972), by Ross (1975), and by Boucot & Gray (1980), as well as the separation of Western Gondwana and Laurussia before and after that time, as first portrayed by Morel & Irving (1978, figs 9, 13).

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